

RQ Answers

omitted

2024-04-05

Data overview

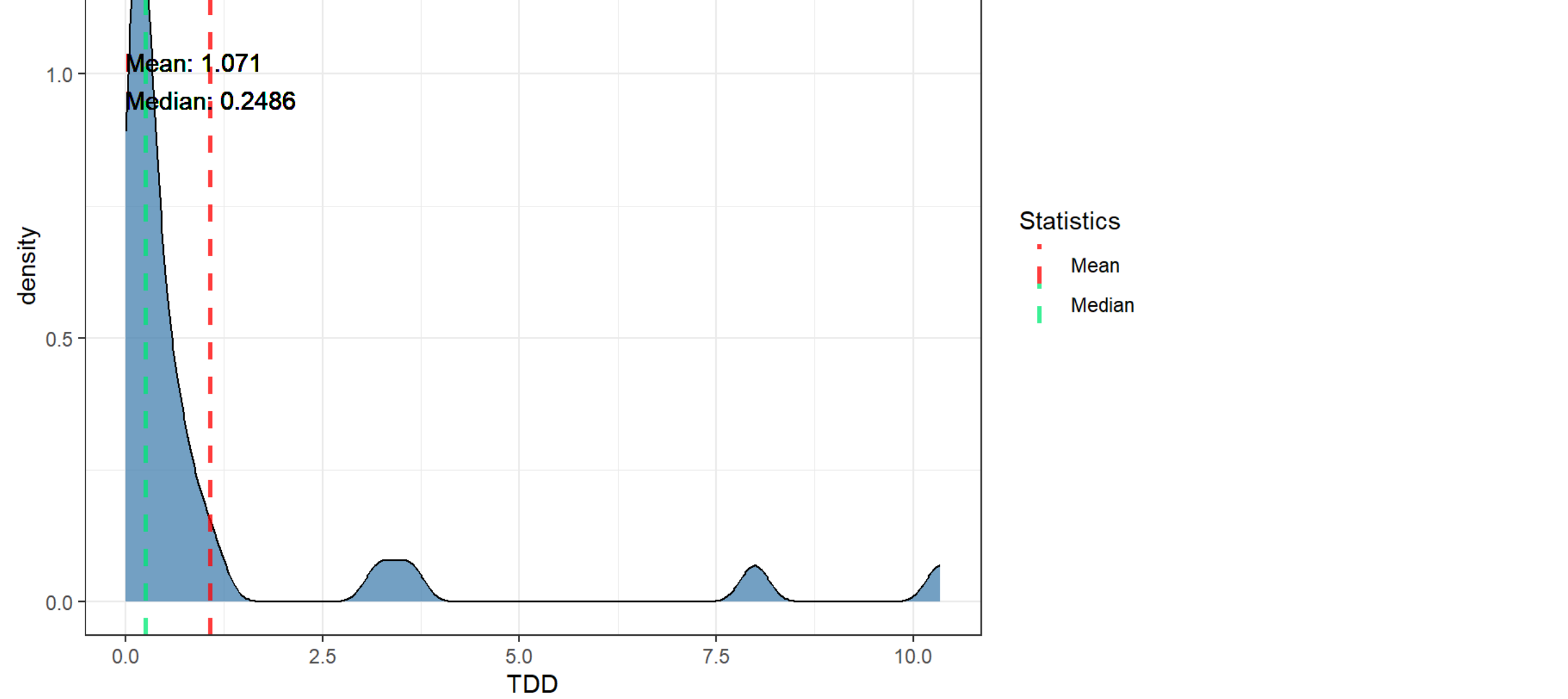
glimpse(all_data)				
## Rows: 780				
## Columns: 11				
## \$ repo	<chr> "wicket", "wicket", "wicket", "wicket", "wicket", ...			
## \$ pr_number	<int> 284, 284, 284, 284, 284, 284, 284, 284, 284, ...			
## \$ rule	<chr> "java:S1161", "java:S110", "java:S5993", "java:S11...",			
## \$ severity	<chr> "MAJOR", "MAJOR", "MAJOR", "MAJOR", "MINOR", "MAJO...",			
## \$ file	<chr> "wicket-extensions/src/main/java/org/apache/wicket...",			
## \$ type	<chr> "CLOSED", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", ...			
## \$ status	<chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...",			
## \$ debt	<int> 5, 270, 2, 330, 1, 10, 1, 5, 1, 2, 5, 5, 5, 2, 2, ...			
## \$ ncloc_affected_file	<int> 46, 46, 46, 46, 475, 475, 475, 475, 475, 475, ...			
## \$ complexity	<int> 7, 7, 7, 7, 79, 79, 79, 79, 79, 79, 79, 79, 79, ...			
## \$ origin	<chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P...			

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the data has 32 rows.
## # Please consider using `annotate()` or provide this layer with data containing
## # a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but the data has 32 rows.
## # Please consider using `annotate()` or provide this layer with data containing
## # a single row.
```

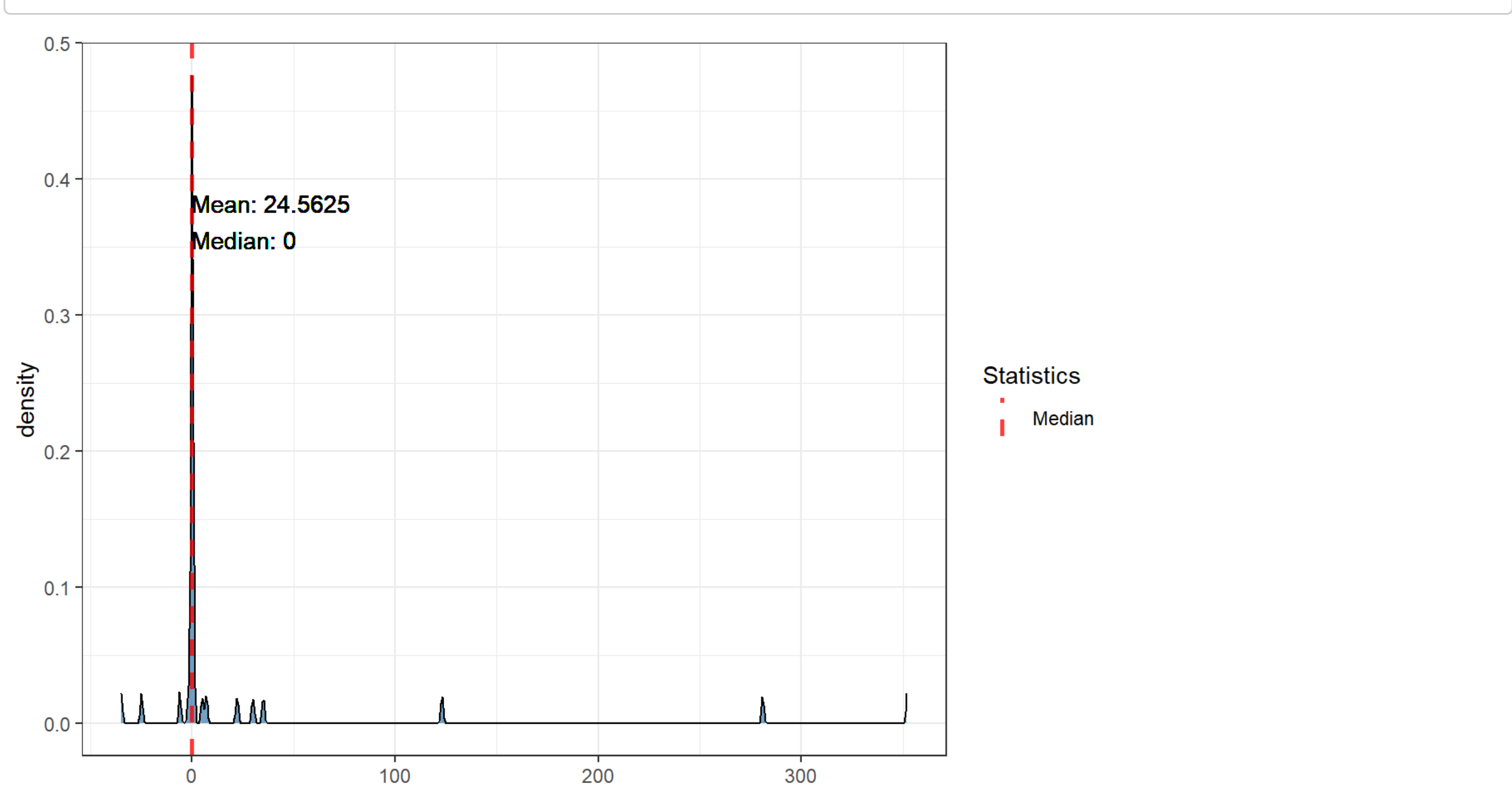


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the data has 32 rows.
## # Please consider using `annotate()` or provide this layer with data containing
## # a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but the data has 32 rows.
## # Please consider using `annotate()` or provide this layer with data containing
## # a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	
## 1 wicket	15.6	59.4	25	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3			
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>	<dbl>
## 1	15.6	59.4	25

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3			
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>
## 1 wicket	61.3	38.7	

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	61.3	38.7

Percentages for new TDV by repo

## # A tibble: 1 × 3			
## # Groups: repo [1]			
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>
## 1 wicket	88.9	11.1	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	88.9	11.1

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank	wicket	
## <int>	<chr>	
## 1	1 java:S1161	- 7
## 2	2 java:S2293	- 7
## 3	3 java:S3776	- 3
## 4	4 java:S1123	- 3
## 5	5 java:S2142	- 2
## 6	6 java:S1124	- 2
## 7	7 java:S3329	- 2
## 8	8 java:S5993	- 1
## 9	9 java:S1488	- 1
## 10	10 java:S1128	- 1

TOP 10 fixed

TOP 10 fixed rules by *PositionScore* metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1161	122	
## 2 java:S2293	123	
## 3 java:S1123	124	
## 4 java:S3776	125	
## 5 java:S1124	126	
## 6 java:S2142	127	
## 7 java:S3329	128	
## 8 java:S1068	129	
## 9 java:S108	130	
## 10 java:S1116	131	

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank	wicket	
## <int>	<chr>	
## 1	1 java:S1125	- 60
## 2	2 java:S110	- 42
## 3	3 java:S1133	- 42
## 4	4 java:S3776	- 41
## 5	5 java:S1066	- 37
## 6	6 java:S1186	- 36
## 7	7 java:S1117	- 33
## 8	8 java:S1604	- 31
## 9	9 java:S2293	- 23
## 10	10 java:S1452	- 20

TOP 10 unfixed

TOP 10 unfixed rules by *PositionScore* metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1125	122	
## 2 java:S110	123	
## 3 java:S1133	124	
## 4 java:S3776	125	
## 5 java:S1066	126	
## 6 java:S1186	127	
## 7 java:S1117	128	
## 8 java:S1604	129	
## 9 java:S2293	130	
## 10 java:S1452	131	

Outliers

Higher TDD

## # A tibble: 32 × 5				
## repo	pr_number	total_debt	total_ncloc	tdd
## <chr>	<int>	<int>	<int>	<dbl>
## 1 wicket	676	600	58	10.3
## 2 wicket	708	1996	250	7.98
## 3 wicket	351	2159	599	3.60
## 4 wicket	495	2125	660	3.22
## 5 wicket	284	7052	6215	1.13
## 6 wicket	385	522	543	0.961
## 7 wicket	428	369	471	0.783
## 8 wicket	462	490	716	0.684
## 9 wicket	588	108	161	0.671
## 10 wicket	357	20	38	0.526
## # i 22 more rows				

Higher TDV

## # A tibble: 32 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	495 wicket	352	
## 2	462 wicket	281	
## 3	351 wicket	123	
## 4	598 wicket	35	
## 5	581 wicket	30	
## 6	470 wicket	22	
## 7	430 wicket	7	
## 8	588 wicket	5	
## 9	285 wicket	0	
## 10	357 wicket	0	
## # i 22 more rows			

Lower TDV

## # A tibble: 32 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	284 wicket	-35	
## 2	708 wicket	-25	
## 3	385 wicket	-6	
## 4	535 wicket	-2	
## 5	709 wicket	-1	
## 6	285 wicket	0	
## 7	357 wicket	0	
## 8	428 wicket	0	
## 9	452 wicket	0	
## 10	453 wicket	0	
## # i 22 more rows			

Extra

Number of PRs with pre-existing TD

n
1 31

Number of java:S1192 issues that affect test code

n
1 0

Number of java:S1192 issues that affect not test code

n
1 18

Summary NCLOC

ncloc
Min. : 29.0
1st Qu.: 173.0
Median : 314.0
Mean : 733.1
3rd Qu.: 827.2
Max. : 6215.0